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PROBLEM & USERS: ECOSYSTEMS ACROSS AN AREA

How could new development affect an area's water supply and rivers through the year?

PROJECT GOAL

Water available each season
Water used and returned
Town reserves + local rivers

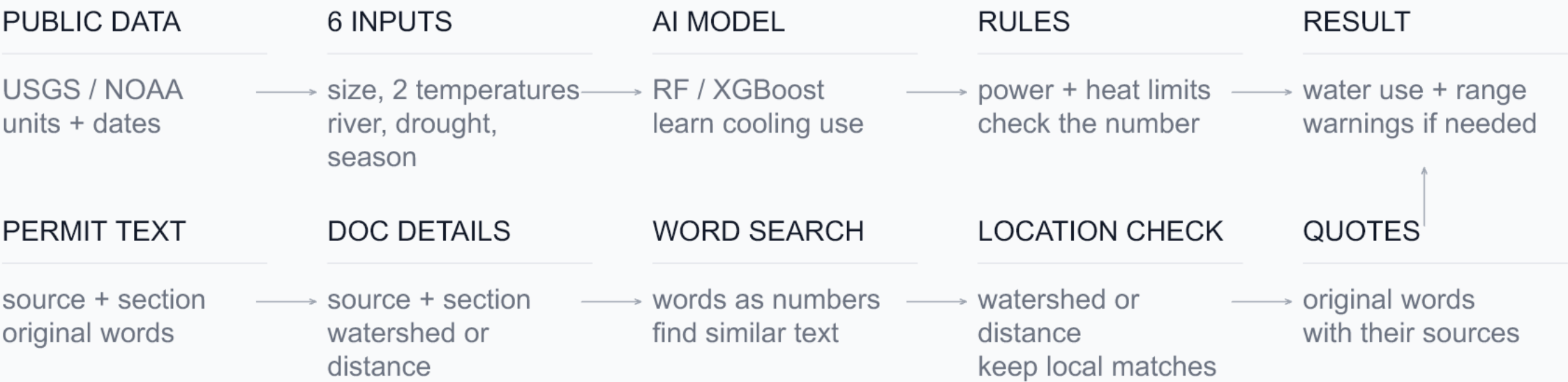
WORKING DEMO

Weather data + cooling-water model
Rules check numbers + permit quotes
Full area assessment still to build

Cooling is one example. Users: town planners, researchers, water providers.

SEMANTIC MAP: CURRENT SYSTEM

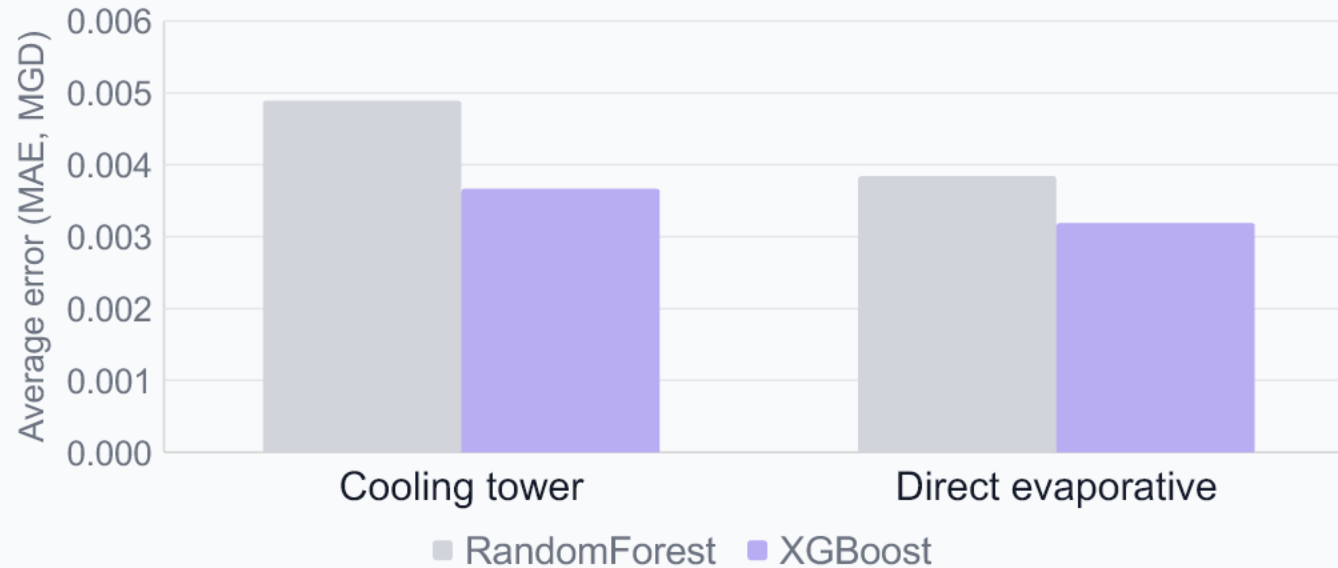
Demo request: equipment size, cooling type, watershed code, month. Exact location is optional.



The AI estimates water use. Rules check it. Local permit quotes explain the context.

E3: forced wrong results stay within 0-0.446484 MGD at 40 MW. E4: all 4 location tests pass.

EXPERIMENT E1: COMPARING AI MODELS



XGBOOST: LOWER ERROR

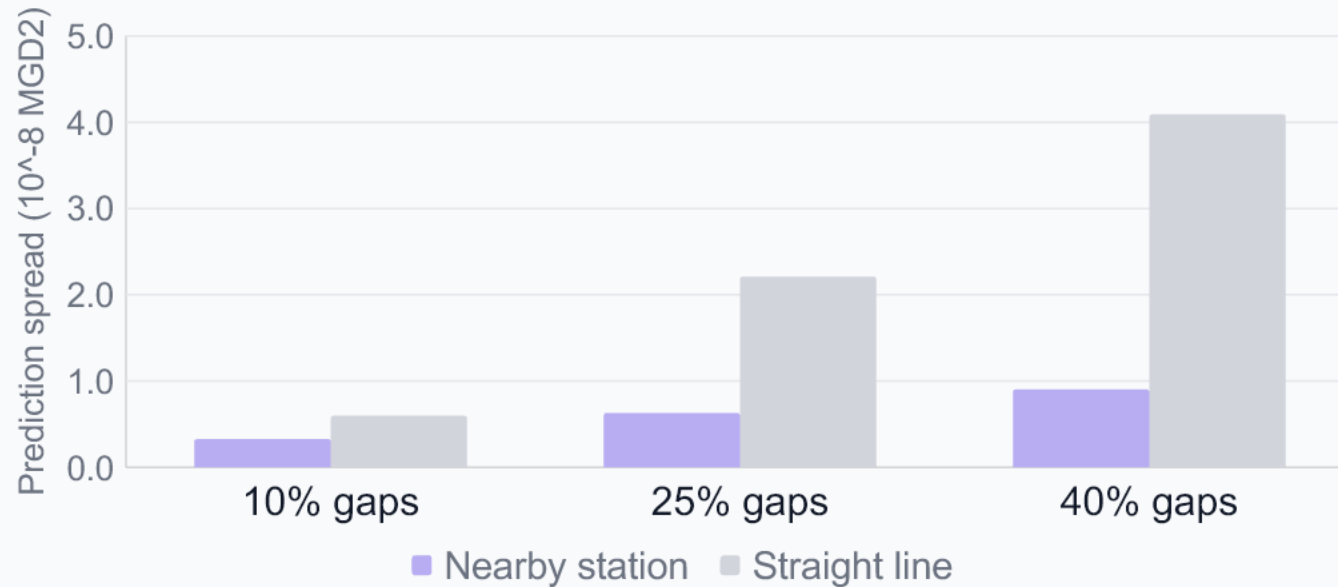
Pick using error on the middle date group. Keep RF to compare.

Next: real water use and area measurements.

Dates: train 60% / choose + set ranges 20% / test 20%

Dry air cooling uses zero water here. Its R2 fit score cannot be calculated.

EXPERIMENT E2: MISSING READINGS



100%

All main tests passed

Change stayed within ± 0.05 MGD
50 repeats on 121 drought days

FAILURE: high readings
only 60.3% within limits

Next: test bad neighbor readings and stations losing data together. The bad-reading test uses a simple formula.

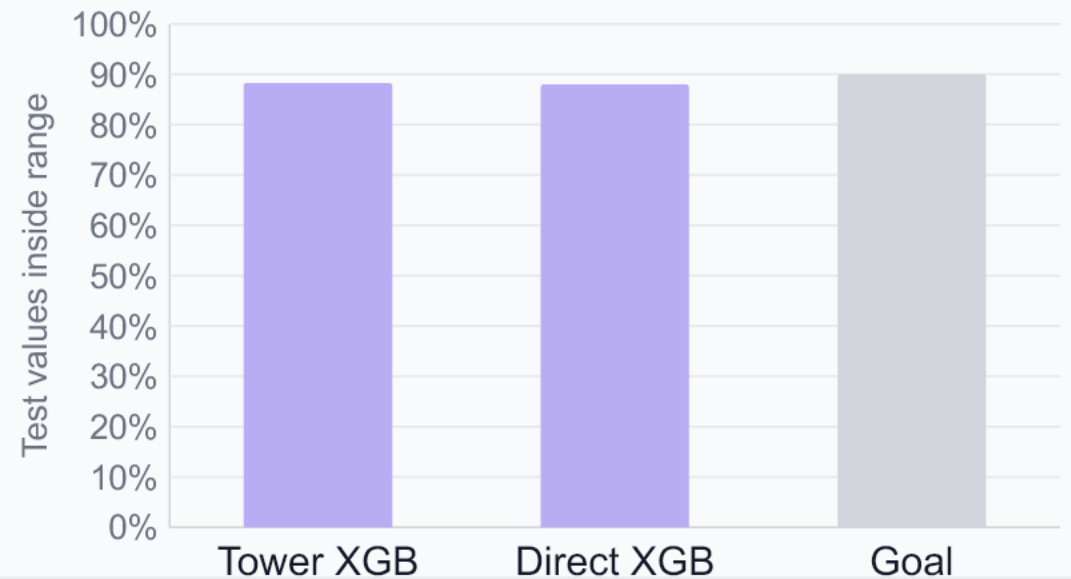
EXPERIMENT E₅: PREDICTION RANGES

88.3% / 88.0%

RELIABILITY ISSUE: below the 90% goal

Tower fit score (R^2) = 0.998

A high score cannot promise a useful range.



Improve: use separate data to choose models and set ranges.

A small test. We have not checked if the shortfall is more than chance. Real-world range tests are still needed.

DECISIONS & NEXT QUESTION

KEEP (ADOPT)

E1: lower XGBoost error. E3/E4: checks pass.
Keep the models, limits, and local matches.

CHANGE (MODIFY)

E2: bad neighbor readings. E5: below 90%.
Improve gap filling and ranges. Add area water totals.

DO NOT USE (REJECT)

E5: a fit score cannot promise a right answer.
Cooling use alone cannot show ecosystem health.

WAIT (DEFER)

Need real area data before testing more AI methods.

Can we test effects on water reserves and rivers in a new area through the year?

Shaun: code + connect area data
Troy: data + source quotes
Scott: tests + results